

The University of Hong Kong

School of Computing and Data Science

COMPXXXX Course Name

Assignment 1: Assignment Title

Author

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Date

6 November 2026

Question 1 Mathematical analysis

(a) Problem formulation

Define the task, assumptions and notation before presenting the solution. For example, let $X \in \mathbb{R}^{n \times d}$ be a design matrix and $\mathbf{y} \in \mathbb{R}^n$ the response vector. Consider the ridge regression objective

$$\mathcal{L}(\mathbf{w}) = \frac{1}{n} \|X\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2, \qquad \lambda > 0. \tag{1}$$

(b) Solution

Differentiating Equation (1) and setting the gradient to zero gives

$$\nabla_{\mathbf{w}} \mathcal{L} = \frac{2}{n} X^\top (X\mathbf{w} - \mathbf{y}) + 2\lambda \mathbf{w}, \tag{2}$$

$$\widehat{\mathbf{w}} = (X^\top X + n\lambda I_d)^{-1} X^\top \mathbf{y}. \tag{3}$$

The positive regularization parameter makes the system matrix positive definite. In code, solve the linear system directly rather than explicitly computing its inverse.

Question 2 Implementation and results

(a) Method

Describe the data, preprocessing, evaluation metric and experimental setup. Use a fixed seed where appropriate, and fit preprocessing steps on training data only. Keep enough detail for the experiment to be reproduced.

```python
import numpy as np

def fit_ridge(X, y, lam=0.1):
    n, d = X.shape
    A = X.T @ X + n * lam * np.eye(d)
    return np.linalg.solve(A, X.T @ y)
```

Listing 1. A minimal ridge regression implementation.

(b) Results and discussion

Table 1 illustrates a simple results table. The values are fictional examples; replace them with measured results and explain the observed differences.

Table 1. Illustrative results only (lower RMSE is better).

Method	Validation RMSE	Test RMSE
Mean baseline	1.24	1.28
Ridge regression	0.87	0.91

(c) Figures

Replace the box below with your own figure. For example:

```
\includegraphics[width=.75\linewidth]{your-figure.pdf}
```

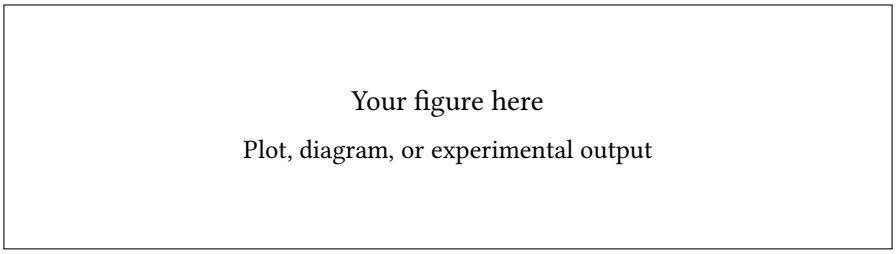


Figure 1. Replace this placeholder and caption with your own figure.

Question 3 Conclusion

Summarize the main finding, explain one limitation, and relate the result to the question. Cite external sources when appropriate, for example the reference repository used for the visual inspiration of this template [1].

References

[1] Guanying Chen. *HKU-PhD-Thesis-LaTex*.
https://github.com/guanyingc/HKU-PhD-Thesis-LaTex.